

Anuk Centellas

anukzc@gmail.com | anukzc.github.io | linkedin.com/in/anuk-centellas

Passionate about applying machine learning and statistical modeling to solve complex, real-world problems.

EDUCATION

University of Washington

MS in Computational Linguistics

Sept 2025 – Present

Expected Graduation June 2027, 4.0 GPA

Western Washington University

BS in Mathematics/Computer Science, BA in Linguistics

Sept 2021 – June 2025

Honors Interdisciplinary Studies Minor, Outstanding Graduate in Linguistics, 4.0 GPA

Coursework: Machine Learning, NLP, Statistics and Probability, Algorithms and Data Structures, Advanced Statistical Methods for NLP

EXPERIENCE

Language Revitalization Lab

Feb 2025 – Present

Documentation, Analysis, and Revitalization in Linguistics (DARLing) - Dr Maura O'Leary Western Washington University

- Analyze and structure under-documented linguistic data for integration into a morphological transducer system
- Improve code consistency and correctness by identifying and correcting errors affecting downstream translation accuracy
- Collaborate with team to determine representations for ambiguous verb morphology, prioritizing the speaker community

Machine Learning Research

April 2024 – June 2025

Hutch Research - Dr Brian Hutchinson

Western Washington University

- Collaborated with the Speech Pathology Department on an applied ML research project to transcribe audio of toddler speech
- Built an end-to-end data preprocessing pipeline in Python to transform noisy transcripts into structured, model-ready datasets
- Independently researched and tested pre-existing forced alignment and diarization models to find building blocks for our model
- Communicated weekly with non-technical stakeholders to ensure our project was maximally helpful for speech pathologists
- Led my team in organizing and assigning tasks, frequent communication, and planning collaborative work sessions

PROJECTS

Data Formatting and Plotting App

Aug 2024 – June 2025

Language Learning Research Group (LLRG)

Python, Streamlit, pandas

- Built and deployed a Streamlit web application enabling users to upload raw Excel data and generate structured outputs
- Collaborated with non-technical team and users to design the app, and independently learned Streamlit to implement it
- Designed and implemented a backend data normalization pipeline using pandas to standardize inconsistent formatting
- Generated exportable visualizations and reformatted data used by the LLRG to improve the Spanish program
- Presented the app and design process to a diverse, non-technical audience as my Honors Senior Capstone Project

Phoneme Classifier

Nov 2024

Deep/Machine Learning Project

Python, Scikit-learn, WandB, GitHub

- Trained and evaluated an ensemble model of twelve neural network classifiers on 1.7 million training instances
- Improved ensemble model performance by giving the top four performing NNs more votes
- Conducted extensive hyperparameter tuning and compared performance across model architectures to find the optimal model
- Achieved a validation accuracy of 0.456 and a test accuracy z-score of 1.340, the top performance for this task

Arbitrarily Deep Neural Network

Oct 2024

Deep/Machine Learning Project

Python, NumPy, argparse, GitHub

- Built a fully customizable neural network framework from scratch through pair programming
- Implemented support for both classification and regression tasks, minibatching, and an optional verbose mode
- Designed the neural network to accept data files and hyperparameters as command-line arguments for flexible experimentation
- Incorporated periodic evaluation on the development set with real-time performance reporting

TECHNICAL SKILLS

Python, Java; NumPy, pandas, Scikit-learn, Matplotlib, argparse, Weights & Biases; Git, GitHub, Linux/Unix, Streamlit; data preprocessing, model evaluation, hyperparameter tuning, experiment tracking, LLM/NLP methods